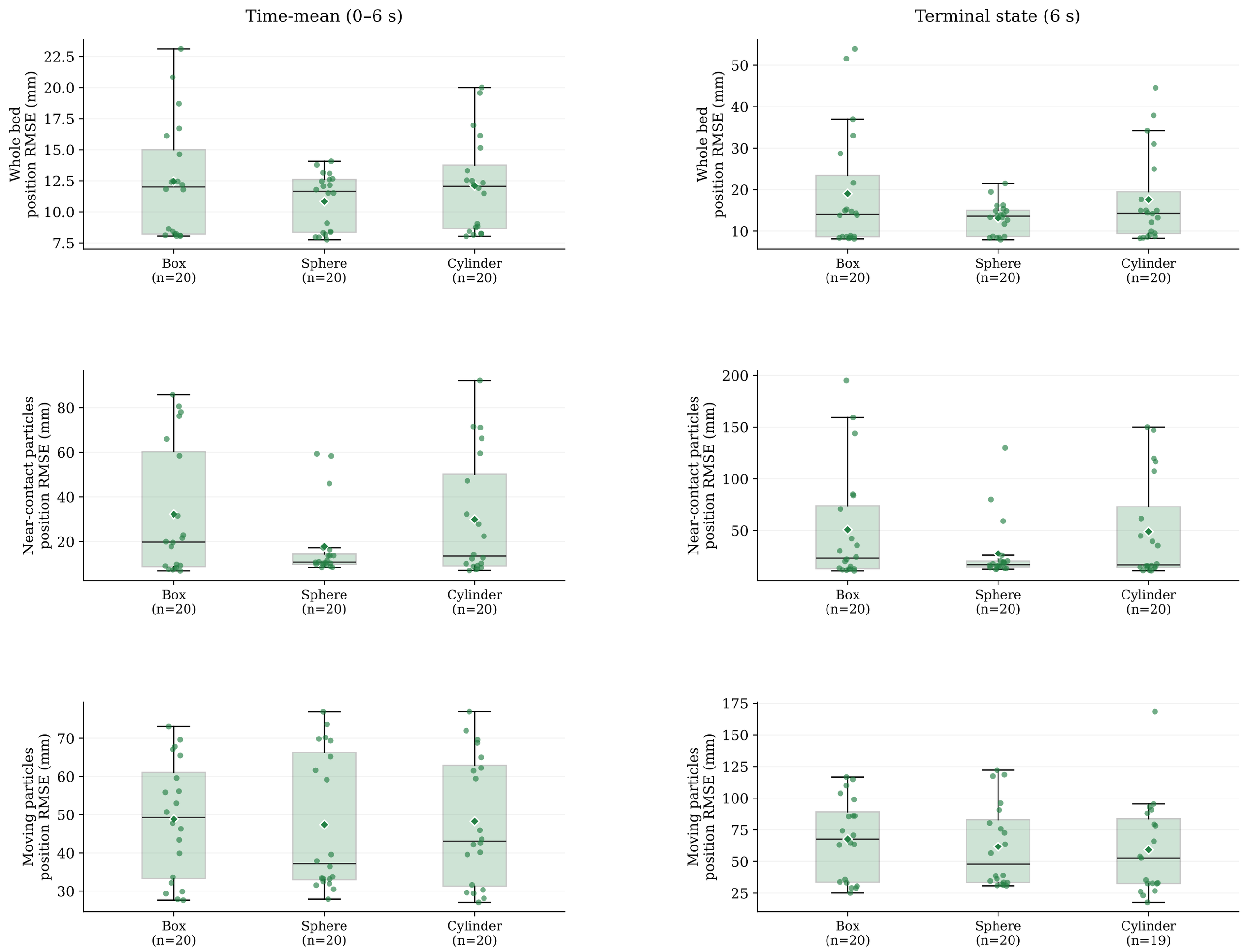

Figure 1: Frozen object-model checkpoint 8,500. Rows show particle-position RMSE for the entire bed, near-contact particles, and moving particles. Left column: time-mean over 0–6 s; right column: terminal state at 6 s. NeRD errors use corresponding MPM particles. Near-contact reference distance ≤0.05 m; moving reference speed >0.01 m/s. One terminal-state moving region is empty and remains unavailable. Each dot is one trajectory summary; boxes show the median and interquartile range, whiskers extend to 1.5 IQR, diamonds mark means. Twenty trajectories per shape unless labeled otherwise. Time means use available samples; empty particle regions are not assigned zero error.